

Stand-Alone Ethernet Controller with SPI Interface

Ethernet Controller Features

- IEEE 802.3™ Compatible Ethernet Controller
- Fully Compatible with 10/100/1000Base-T Networks
- Integrated MAC and 10Base-T PHY
- Supports One 10Base-T Port with Automatic Polarity Detection and Correction
- Supports Full and Half-Duplex modes
- Programmable Automatic Retransmit on Collision
- Programmable Padding and CRC Generation
- Programmable Automatic Rejection of Erroneous Packets
- SPI Interface with Clock Speeds Up to 20 MHz

Buffer

- 8-Kbyte Transmit/Receive Packet Dual Port SRAM
- Configurable Transmit/Receive Buffer Size
- Hardware Managed Circular Receive FIFO
- Byte-Wide Random and Sequential Access with Auto-Increment
- Internal DMA for Fast Data Movement
- Hardware Assisted Checksum Calculation for Various Network Protocols

Medium Access Controller (MAC) Features

- Supports Unicast, Multicast and Broadcast Packets
- Programmable Receive Packet Filtering and Wake-up Host on Logical AND or OR of the Following:
 - Unicast destination address
 - Multicast address
 - Broadcast address
 - Magic Packet™
 - Group destination addresses as defined by 64-bit Hash Table
 - Programmable Pattern Matching of up to 64 bytes at user-defined offset

Physical Layer (PHY) Features

- Loopback mode
- Two Programmable LED Outputs for LINK, TX, RX, Collision and Full/Half-Duplex Status

Operational

- Six Interrupt Sources and One Interrupt Output Pin
- 25 MHz Clock Input Requirement
- Clock Out Pin with Programmable Prescaler
- Operating Voltage of 3.1V to 3.6V (3.3V typical)
- 5V Tolerant Inputs
- Temperature Range: -40°C to +85°C Industrial, 0°C to +70°C Commercial (SSOP only)
- 28-Pin SPDIP, SSOP, SOIC, QFN Packages

Package Types

